STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-DGM-ABELIAN

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1443-1464 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 1466-1509, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 14 of queue_sdga.json (pb=1945, stmt 1443-1464, proof 1466-1509).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-dgm-abelian' also exists in stacks-dga — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 002P: sources/stacks-sdga/pinned-original-source.tex, source lines 1509–1509. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 002N: sources/stacks-sdga/pinned-original-source.tex, source lines 1509–1509. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FRI: sources/stacks-sdga/pinned-original-source.tex, source lines 1396–1418. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FR0: sources/stacks-sdga/pinned-original-source.tex, source lines 386–397. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a differential graded $\mathcal{O}$-algebra. The category $Mod(\mathcal{A}, d)$ is an abelian category with the following properties*

(1) *$Mod(\mathcal{A}, d)$ has arbitrary direct sums,*

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *$Mod(\mathcal{A}, d)$ has arbitrary colimits,*
(3) *filtered colimit in $Mod(\mathcal{A}, d)$ are exact,*
(4) *$Mod(\mathcal{A}, d)$ has arbitrary products,*
(5) *$Mod(\mathcal{A}, d)$ has arbitrary limits.*

The forgetful functor

$$Mod(\mathcal{A}, d) \longrightarrow Mod(\mathcal{A})$$

sending a differential graded $\mathcal{A}$-module to its underlying graded module commutes with all limits and colimits.

Proof. Let us denote $F : Mod(\mathcal{A}, \mathrm{d}) \to Mod(\mathcal{A})$ the functor in the statement of the lemma. Observe that the category $Mod(\mathcal{A})$ has properties (1) – (5), see Lemma 0FR0.

Consider a homomorphism $f : \mathcal{M} \to \mathcal{N}$ of graded $\mathcal{A}$-modules. The kernel and cokernel of f as maps of graded $\mathcal{A}$-modules are additionally endowed with differentials as in Definition 0FRI. Hence these are also the kernel and cokernel in $Mod(\mathcal{A}, \mathrm{d})$. Thus $Mod(\mathcal{A}, \mathrm{d})$ is an abelian category and taking kernels and cokernels commutes with F.

To prove the existence of limits and colimits it is sufficient to prove the existence of products and direct sums, see Categories, Lemmas 002N and 002P. The same lemmas show that proving the commutation of limits and colimits with F follows if F commutes with direct sums and products.

Let $\mathcal{M}_t$, $t \in T$ be a set of differential graded $\mathcal{A}$-modules. Then we can consider the direct sum $\bigoplus \mathcal{M}_t$ as a graded $\mathcal{A}$-module. Since the direct sum of graded modules is done termwise, it is clear that $\bigoplus \mathcal{M}_t$ comes endowed with a differential. The reader easily verifies that this is a direct sum in $Mod(\mathcal{A}, \mathrm{d})$. Similarly for products.

Observe that F is an exact functor and that a complex $\mathcal{M}_1 \to \mathcal{M}_2 \to \mathcal{M}_3$ of $Mod(\mathcal{A}, \mathrm{d})$ is exact if and only if $F(\mathcal{M}_1) \to F(\mathcal{M}_2) \to F(\mathcal{M}_3)$ is exact in $Mod(\mathcal{A})$. Hence we conclude that (3) holds as filtered colimits are exact in $Mod(\mathcal{A})$. □